

Client Proof Bundle Report

Production Readiness: ready-production

This report summarizes the latest NightSeal PQC proof chain, including manifest integrity, Merkle proof, ML-DSA signature verification, manifest chain status and client deliverable context.

Executive Summary

READINESS ready-production	READINESS SCORE 100.0%	CRYPTO PROVIDER ml-dsa-65	LATEST ML-DSA VERIFICATION YES
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Report ID	client-report-20260518T194621Z
Generated at	2026-05-18T19:46:21.463364+00:00
Bundle ID	proof-bundle-20260518T192851Z
Bundle status	complete-demo-proof-chain
Source type	oracle-rman

Production Readiness Assessment

Assessment ID	production-readiness-20260518T194621Z
Checked at	2026-05-18T19:46:21.463315+00:00
Readiness	ready-production
Score	100.0%
Passed checks	10
Failed checks	0
Critical failures	0
Recommendation	System is production-ready when ML-DSA is active and the latest ML-DSA proof verifies successfully.

Readiness Checks

Check	Category	Severity	Passed	Details
Backend API running	platform	critical	YES	FastAPI backend is reachable.
Dataset accessible	data	critical	YES	Dataset path is accessible and contains files.

Check	Category	Severity	Passed	Details
Proof chain functional	proof-chain	critical	YES	Manifest, Merkle proof, signature proof and bundle verification are functional.
Manifest chain valid	audit	high	YES	Temporal manifest chaining is valid.
Production crypto provider	crypto	critical	YES	A real ML-DSA provider must replace HMAC-SHA256-DEMO before production.
Post-quantum signature enabled	crypto	critical	YES	Current provider must be ML-DSA for production-grade PQC proof.
Latest signature uses ML-DSA	crypto-verification	critical	YES	The latest signature proof must have been generated with ML-DSA.
Latest ML-DSA signature verifies	crypto-verification	critical	YES	The latest ML-DSA signature must verify cryptographically against its payload.
Client reporting available	reporting	medium	YES	HTML client report generation is available.
Smoke-testable deployment	operations	medium	YES	PowerShell smoke test is available in tools/nightseal-smoke-test.ps1.

ML-DSA Signature Verification

Signature proof	signature-proof-20260518T192851Z.json
Signature mode	ML-DSA-65
Verification status	valid
Valid	YES
Payload SHA-256	b5320f60a7d332ed8d5feab740dd6903d67a54c4102398bf41b69ef40a2bb4ed
Key source	/data/keys/ml-dsa-65.secret.key

Crypto Provider Status

Active provider	ml-dsa-65
Crypto status	production-ready
Production ready	YES
Available providers	hmac-demo, ml-dsa-44, ml-dsa-65, ml-dsa-87

Latest Workflow Run

Workflow ID	full-proof-chain-20260518T192851Z
Created at	2026-05-18T19:28:51.977429+00:00
Status	success
Verification valid	YES
Manifest ID	manifest-20260518T192848Z-c46dc2be
Bundle ID	proof-bundle-20260518T192851Z

Manifest Chain

Status	valid
Valid	YES
Manifest count	30
Linked manifest count	20

Manifest

Filename	manifest-20260518T192848Z-c46dc2be.json
Manifest ID	manifest-20260518T192848Z-c46dc2be
File count	6
Total size bytes	399507456
Hash algorithm	SHA-256
Manifest SHA-256	5f62f5dba088419add783b1f6d990dd9ad7fd2d4a494305261aecdda237ac7ce

Merkle Proof

Filename	merkle-proof-20260518T192851Z.json
Proof ID	merkle-proof-20260518T192851Z
Leaf count	6
Tree height	4
Merkle root	ac0764cf9755717b6049805da34da69caa83c74e901ea910c47d26996f938d7a

Merkle proof SHA-256	2829a96a2356c98434833e3a94b86fdff14859f3243f159b7c45be143f20912d
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Signature Proof in Bundle

Filename	signature-proof-20260518T192851Z.json
Signature proof ID	signature-proof-20260518T192851Z
Signature mode	ML-DSA-65
Payload SHA-256	b5320f60a7d332ed8d5feab740dd6903d67a54c4102398bf41b69ef40a2bb4ed
Status	real-pqc-signature

Security Note

NightSeal PQC now reports production readiness only when a real ML-DSA provider is active and the latest ML-DSA signature verifies successfully. Storage immutability and cryptographic proof complement each other: WriteOnce protects against modification, while NightSeal PQC proves integrity and auditability.